Pseudonymized but Linkable: Cross-Document Re-identification Risks in LLM-Ready Sensitive Text Corpora

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Abstract

Sensitive text corpora are commonly de-identified one document at a time before retrieval, evaluation, or model adaptation. We study a corpus-level failure mode for LLM-ready data: records with no direct identifiers can remain linkable through repeated quasi-identifiers such as role, region, schedule, institution type, family context, health context, and rare events. We introduce CrossDoc-PrivacyBench, a fully synthetic benchmark with 120 controlled personas, 480 documents, risk tiers, utility labels, and decoy auxiliary profiles. Direct redaction leaves auxiliary-profile matching unchanged at 0.708 top-1, while Presidio-style PII redaction remains highly matchable at 0.594. We propose LinkGuard, a simple corpus-aware generalization heuristic. At target $k = 5$, LinkGuard reduces auxiliary top-1 to 0.042 while preserving issue classification and retrieval utility at 1.000; aggressive redaction reaches similar privacy but drops issue accuracy to 0.350 and retrieval Recall@5 to 0.312. These synthetic results suggest that LLM-ready data transformations should be evaluated for corpus-level linkage resistance, not only document-local PII removal.

1 Introduction

Sensitive records are commonly transformed one document at a time: remove names, emails, phone numbers, addresses, account numbers, and perhaps other obvious personal identifiers. This is operationally convenient for RAG databases, internal analytics, evaluation sets, and model-adaptation corpora, but it ignores the fact that many corpora contain several documents per person or case. Weak details that look harmless in isolation can become a stable signature once repeated across healthcare, legal, financial, and workplace records.

Prior work shows that text anonymization should be evaluated by residual re-identification risk rather than span recall alone (Krčo et al., 2026), and that language models can infer private attributes from text (Staab et al., 2024). RAG privacy work similarly warns that retrieval databases can leak sensitive records (Zeng et al., 2024). Our differentiating question is narrower and complementary: after document-local anonymization, can an attacker still link multiple pseudonymized documents to the same synthetic person and match them to an auxiliary profile?

We make three contributions. First, we introduce CrossDoc-PrivacyBench, a synthetic cross-document privacy benchmark with controlled ground truth and no real-person data. Second, we evaluate linkage, auxiliary-profile matching, risk-tier behavior, RAG-style exposure, and utility retention across common transformations. Third, we test LinkGuard, a corpus-aware heuristic that generalizes high-linkage quasi-identifiers according to corpus-level uniqueness.

2 Benchmark and threat model

The data holder has a multi-document sensitive text corpus and transforms it before indexing, sharing, or using it in foundation-model workflows. The attacker sees the transformed corpus. For the auxiliary-profile task, the attacker also sees a ten-candidate synthetic profile set containing the true persona plus decoys that share some fields such as region, occupation family, or legal context. The attacker's goals are to link records from the same synthetic persona and rank the correct auxiliary profile. All personas, documents, profiles, and decoys are generated by the experimenter; we use no real people, public profiles, social-media posts, patient records, legal records, or customer records.

CrossDoc-PrivacyBench contains 120 personas and 480 documents. Each persona has direct identifiers, utility labels, and quasi-identifiers: region, city, occupation, employer type, education, family structure, medical context, financial context, legal context, schedule pattern, affiliation, and rare event. Each persona has four documents spanning healthcare, legal, financial support, and HR. Personas are assigned to T1/T2/T3 risk tiers, where T1 mostly contains direct PII with generic context, T2 contains moderate quasi-identifiers, and T3 contains high-linkage combinations. We evaluate 96 held-out personas.

3 Transformations and metrics

We compare original documents, direct redaction, Presidio-based PII redaction, stable pseudonyms, per-document pseudonyms, a document-local anonymization proxy, LinkGuard, and aggressive redaction. LinkGuard estimates cross-document linkage risk by scoring quasi-identifier fields by rarity, specificity, stability, and approximate utility cost. It then moves fields from exact value to semantic category and, if needed, to a generic suppression phrase until the persona representation reaches a target anonymity set size. This is a deliberately simple heuristic, not a formal privacy guarantee.

Privacy metrics include pairwise linkage F1, fixed-K clustering ARI, auxiliary-profile top-1/top-3/MRR, and exact quasi-identifier recovery. The main local auxiliary matcher uses word TF-IDF; robustness checks add character, hybrid, and field-weighted matchers. The field-weighted matcher explicitly scans exact and generalized schema fields, making it a stress test for residual structured context. Utility metrics include domain/issue classification, task-oriented retrieval Recall@5, and lightweight fact preservation.

4 Results

The main sweep is summarized below. Direct redaction leaves auxiliary matching unchanged at 0.708 top-1, confirming that direct PII removal alone does not remove repeated quasi-identifier risk. Presidio-style PII redaction improves but remains highly matchable at 0.594. Consistent pseudonymization makes linkage nearly trivial, with pairwise F1 0.979. The document-local proxy still leaves auxiliary top-1 at 0.604 and exact quasi-identifier recovery at 0.304.

Cond.	Pair	Aux@1	Exact	Issue	Ret@5	Facts	Edit
C0 Orig	0.963	0.708	0.395	1.000	1.000	0.682	0.000
C1 Redact	0.214	0.708	0.395	1.000	1.000	0.682	0.246
C1b Presidio	0.142	0.594	0.332	1.000	1.000	0.682	0.348
C2 Stable	0.979	0.427	0.273	1.000	1.000	0.679	0.299
C3 PerDoc	0.148	0.427	0.273	1.000	1.000	0.679	0.315
C4 Local	0.155	0.604	0.304	1.000	1.000	0.682	0.288
C5 LG	0.019	0.042	0.000	1.000	1.000	0.695	0.535
C6 Agg	0.016	0.052	0.000	0.350	0.312	0.389	0.681

LinkGuard reduces auxiliary top-1 to 0.042 and exact recovery to 0.000 while preserving issue accuracy and retrieval Recall@5 at 1.000. Aggressive redaction reaches a similar auxiliary top-1 of 0.052, but issue accuracy drops to 0.350 and retrieval Recall@5 to 0.312. Thus, in this synthetic sprint, corpus-aware generalization occupies a better privacy–utility point than blanket suppression. A stricter body-only utility stress test tells the same story: LinkGuard scores 0.880, direct redaction 0.870, and aggressive redaction 0.438.

The benchmark tiers behave as intended. Under direct redaction, Aux@1 rises from 0.156 on T1 to 0.969 on T2 and 1.000 on T3. LinkGuard lowers the corresponding rates to 0.031, 0.000, and 0.094. Bootstrap 95% confidence intervals over held-out personas keep the main contrast separated: direct redaction 0.708 [0.615, 0.802], Presidio 0.594 [0.490, 0.698], and LinkGuard 0.042 [0.010, 0.083]. A three-seed no-API sweep preserves the ordering: direct redaction, Presidio, and the document-local proxy have mean Aux@1 0.719, 0.628, and 0.663, while LinkGuard averages 0.045. A noisy-style synthetic rerendering lowers mean template similarity to 0.306 and preserves the ordering: direct/Presidio/document-local Aux@1 0.656/0.562/0.594, LinkGuard 0.042 with issue/Ret@5 1.000, and aggressive redaction issue/Ret@5 0.473/0.635.

We also test harder local attackers. Across word, character, hybrid TF-IDF, and field-weighted matchers, direct redaction ranges from 0.510 to 0.979 top-1 and the document-local proxy from 0.479 to 0.979. LinkGuard remains lower, from 0.031 to 0.240, but the field-weighted result exposes residual structured-context risk. The sensitivity sweep below shows that this residual is controllable: field-weighted top-1 falls from 0.979 at $k = 1$ to 0.240 at $k = 5$ and 0.104 at $k = 20$, with issue and retrieval utility still 1.000.

k	Min k	Edit	Pair	Aux@1	Field@1	Exact	Issue	Ret@5
1	1	0.246	0.214	0.708	0.979	0.395	1.000	1.000
2	2	0.521	0.036	0.062	0.354	0.007	1.000	1.000
3	3	0.528	0.029	0.052	0.333	0.004	1.000	1.000
5	5	0.535	0.019	0.042	0.240	0.000	1.000	1.000
8	8	0.538	0.017	0.042	0.167	0.000	1.000	1.000
12	12	0.539	0.017	0.042	0.125	0.000	1.000	1.000
20	20	0.544	0.017	0.042	0.104	0.000	1.000	1.000

For a RAG-style exposure diagnostic, exact synthetic profile queries retrieve at least one target document in the top five for every high-linkage T3 persona under direct redaction, Presidio, and the document-local proxy. LinkGuard reduces T3 Hit@5 to 0.312 and Multi@10 to 0.000. A small cached OpenAI audit on 12 T2/T3 personas is consistent with the local pattern: direct redaction is 0.750 top-1, OpenAI document-local anonymization is 0.667, LinkGuard is 0.250, and aggressive redaction is 0.167.

5 Limitations and ethics

The benchmark is synthetic and template-controlled, which gives safe ground truth but limits external validity. Real corpora may differ in style, temporal structure, institution-specific vocabulary, and adversary knowledge. The local attacks are simple, and the OpenAI audit is intentionally small. LinkGuard is not a formal anonymity guarantee, and the field-weighted attacker shows that residual generalized context can still be informative. Utility is measured with issue labels, retrieval, and lightweight fact checks rather than full downstream deployment tasks.

The benchmark is intended for defensive auditing. We do not collect, process, or attempt to re-identify real people. Any extension to real data would require authorization, data minimization, and ethical review.

6 Conclusion

Document-local PII removal is not enough for LLM-ready sensitive corpora. In a multi-document setting, repeated quasi-identifiers can support linkage and auxiliary-profile matching even after direct identifiers are removed. Our synthetic benchmark suggests that privacy evaluation should include corpus-level linkage resistance, and that simple corpus-aware generalization can improve the privacy–utility frontier relative to document-local anonymization and blanket redaction.

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